AI Powered Anomaly Detection in Medical Imagery

WiDS 5.0 – Final Report

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Abstract

Artificial Intelligence (AI) has emerged as a transformative force in modern healthcare, particularly in the domain of medical imaging. This report documents my learning journey through WiDS 5.0, focusing on the application of computer vision and deep learning techniques to healthcare problems. The project progresses from understanding foundational concepts such as medical imaging reports and digital image processing to implementing convolutional neural networks and YOLOv8-based image classification models. Through theoretical study, hands-on experimentation, and practical challenges, this work highlights how AI can assist clinicians by improving diagnostic accuracy, efficiency, and scalability while also addressing ethical and technical limitations inherent in healthcare AI systems.

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1. Introduction

Healthcare systems across the world generate massive amounts of data every day, much of which comes in the form of medical images such as X-rays, CT scans, MRIs, ultrasounds, and histopathology slides. Interpreting these images requires expert knowledge and significant time, making the diagnostic process resource-intensive and prone to human variability. Artificial Intelligence, particularly machine learning and deep learning, has shown great promise in addressing these challenges.

The Winters in Data Science (WiDS) 5.0 program provided an opportunity to explore how AI techniques can be applied to real-world healthcare problems. This project focuses on AI-powered medical image analysis, combining theoretical understanding with practical implementation. The objective was not only to train models, but also to understand the broader healthcare context, data limitations, ethical responsibilities, and real-world deployment challenges.

This report presents a structured account of all concepts learned, experiments performed, and insights gained throughout the WiDS journey.

2. AI in Healthcare: An Overview

Artificial Intelligence in healthcare refers to the use of algorithms and data-driven models to assist in diagnosis, treatment planning, patient monitoring, and administrative tasks. AI systems can analyse large volumes of data faster than humans and uncover patterns that may not be immediately visible to clinicians.

2.1 Applications of AI in Healthcare

- Disease diagnosis and early detection
- Medical imaging analysis
- Drug discovery and development
- Personalized treatment planning
- Predictive analytics for patient outcomes
- Hospital workflow optimization

Among these, medical imaging has become one of the most successful and impactful applications of AI due to the visual nature of the data and the effectiveness of deep learning models in image understanding.

2.2 Importance of Medical Imaging

Medical imaging plays a crucial role in diagnosing diseases, monitoring treatment response, and planning surgeries. Accurate interpretation of these images can significantly influence patient outcomes. However, the increasing volume of imaging data has created a growing workload for radiologists and pathologists, making automation and AI assistance increasingly necessary.

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3. Healthcare Reports and Medical Imaging

Understanding healthcare data requires familiarity with different types of medical reports and imaging modalities.

3.1 Types of Healthcare Reports

- **Clinical Reports:** Doctor consultation notes, diagnosis summaries, and treatment plans
- **Laboratory Reports:** Blood tests, urine tests, pathology examinations
- **Radiology Reports:** Interpretations of X-rays, CT scans, and MRIs
- **Pathology Reports:** Microscopic analysis of tissue samples, critical for cancer diagnosis
- **Surgical Reports:** Detailed documentation of surgical procedures
- **Preventive and Screening Reports:** Routine health checkups and screenings

3.2 Medical Imaging Modalities

- **X-ray:** Detects fractures and lung-related conditions
- **CT Scan:** Provides detailed cross-sectional images
- **MRI:** High-resolution imaging of soft tissues
- **Ultrasound:** Uses sound waves for internal visualization
- **PET Scan:** Measures metabolic activity, commonly used in oncology

Medical imaging reports follow a structured format that includes patient information, scan details, findings, impressions, and recommendations. AI systems aim to assist clinicians by improving consistency, speed, and accuracy in interpreting these images.

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4. Digital Image Processing Fundamentals

Before applying deep learning models, it is essential to understand how images are represented and processed digitally.

4.1 Image Representation

Digital images are represented as matrices of pixel values. Each pixel stores intensity information, and color images typically use three channels—Red, Green, and Blue (RGB).

4.2 Grayscale Conversion

Converting color images to grayscale reduces computational complexity while retaining essential structural information. Grayscale images are widely used in medical imaging where color information may not be critical.

4.3 Thresholding

Thresholding is a technique used to separate objects from the background by converting grayscale images into binary images. It is commonly used in segmentation and preprocessing pipelines.

4.4 Image Resizing and Interpolation

Medical images often need to be resized for model compatibility. Interpolation methods such as bilinear and bicubic interpolation affect image smoothness and quality. Bicubic interpolation produces smoother results but is computationally more expensive.

4.5 Color Spaces

While RGB is commonly used, HSV (Hue, Saturation, Value) is often more intuitive for image analysis tasks. HSV separates color information from intensity, making it useful for color-based segmentation.

Through hands-on notebooks, these preprocessing techniques were implemented and analyzed to understand their impact on downstream machine learning models.

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5. Convolutional Neural Networks (CNNs)

Convolutional Neural Networks form the backbone of modern computer vision systems. CNNs automatically learn hierarchical features from images, starting from simple edges to complex patterns.

5.1 Key Components of CNNs

- Convolutional layers
- Activation functions
- Pooling layers
- Fully connected layers

5.2 ImageNet and Transfer Learning

ImageNet is a large-scale dataset that enabled significant advancements in deep learning. Many CNN architectures pretrained on ImageNet can be reused for medical imaging tasks through transfer learning, reducing the need for large labeled datasets.

5.3 Popular CNN Architectures

- **AlexNet:** Demonstrated the power of deep CNNs
- **VGG:** Introduced deeper networks with simple architectures
- **ResNet:** Solved vanishing gradient problems using residual connections
- **EfficientNet:** Balanced scaling of depth, width, and resolution

Understanding these architectures provided a strong theoretical foundation for practical model implementation.

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6. YOLO and the Darknet Framework

YOLO (You Only Look Once) is a family of deep learning models designed for real-time object detection and classification.

6.1 Darknet Framework

Darknet is an open-source neural network framework written in C and CUDA, optimized for speed and efficiency. It was the foundation for early YOLO models.

6.2 YOLO Modes

- Train
- Validation
- Predict
- Export
- Track

6.3 YOLO Tasks

- Object detection
- Image classification
- Image segmentation
- Pose estimation

YOLOv8, developed by Ultralytics, extends these capabilities with improved performance and ease of use.

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7. YOLOv8 Image Classification Experiment

7.1 Objective

The goal of this experiment was to gain practical experience in training, validating, and evaluating a YOLOv8 classification model.

7.2 Dataset Exploration

A skin disease dataset from Roboflow was initially explored to understand real-world medical data challenges. However, due to dataset structure and compatibility issues in Google Colab, CIFAR-10 was used for final implementation to ensure a stable training pipeline.

7.3 Model Configuration

- Model: YOLOv8 Nano (Classification)
- Framework: Ultralytics YOLOv8
- Platform: Google Colab (GPU)
- Epochs: 10
- Image size: 224 × 224

7.4 Training and Results

The model showed a consistent decrease in training loss and improvement in classification accuracy over epochs. Sample predictions demonstrated successful inference, confirming the effectiveness of the training process.

8. Automated Detection of Tumor-Infiltrating Lymphocytes using YOLOv8

8.1 Problem Statement

Tumor-Infiltrating Lymphocytes (TILs) are immune cells present within and around tumor regions in histopathology slides. The presence and density of TILs serve as a crucial biomarker in breast cancer, as higher levels of TILs are associated with improved patient prognosis and better response to immunotherapy, particularly in **HER2-positive** and **Triple-Negative Breast Cancer** cases.

Traditionally, TIL scoring is performed manually by pathologists using Hematoxylin and Eosin (H&E) stained whole-slide images. This manual process is:

- Time-consuming
- Subjective
- Difficult to scale
- Prone to inter-observer variability

Whole-slide images are extremely large and complex, making consistent and reproducible scoring challenging. To address these limitations, AI-based methods are being explored for automated detection and quantification of TILs.

This project aligns with the **TIGER Grand Challenge**, which aims to automate lymphocyte and plasma cell detection, tumor and stroma segmentation, and ultimately compute automated TIL scores. The focus of this Week 4 task is the **automatic detection of lymphocytes and plasma cells using deep learning**.

8.2 Importance of the Problem

Breast cancer is one of the most common cancers worldwide and a leading cause of cancer-related mortality among women. Accurate assessment of TILs can help clinicians:

- Predict patient survival
- Identify patients likely to benefit from immunotherapy
- Reduce unnecessary chemotherapy
- Personalize treatment strategies

However, the shortage of trained pathologists and the growing volume of medical imaging data make manual analysis unsustainable. AI-driven detection systems offer:

- Faster diagnosis
- Consistent and reproducible scoring
- Scalable solutions for healthcare systems
- Improved clinical decision-making

Projects such as TIGER and AIRAT demonstrate that AI in histopathology is transitioning from research to real-world clinical deployment.

8.3 Dataset Description

For this task, the **TIGER WSIRois dataset** was used, obtained via Roboflow. The dataset contains:

- Histopathology image patches
- Bounding box annotations for lymphocytes and plasma cells
- Labels provided in COCO annotation format

Due to the extremely large size of whole-slide images, a **patch-based learning approach** was adopted. This allows efficient training while preserving clinically relevant information.

8.4 Methodology and Pipeline

A complete object detection pipeline was designed using **YOLOv8**.

Pipeline Overview

1. **Dataset Preparation**
    - Use annotated histopathology image patches
    - Convert annotations from COCO format to YOLO format
    - Resize images to 512 × 512 for consistent training
2. **Preprocessing**
    - Image resizing and normalization
    - Data augmentation to improve generalization
    - Train-validation split
3. **Model Architecture**
    - YOLOv8 object detection network
    - Pretrained backbone for transfer learning

YOLOv8 was chosen due to its efficiency, speed, and strong performance in detecting small objects, which is critical for lymphocyte detection.

8.5 Challenges and Observations

This task highlighted several real-world challenges unique to medical imaging:

- **Small Object Size:**
  Lymphocytes are extremely small and densely packed, making detection significantly harder than general object detection tasks.
- **Stain Variability:**
  Differences in staining across histopathology slides affect color consistency and model generalization.
- **Class Imbalance:**
  Lymphocytes appear far more frequently than plasma cells, leading to imbalance issues during training.
- **Large Image Size:**
  Whole-slide images are computationally expensive, necessitating patch-based learning.
- **Overfitting Risk:**
  Medical datasets are often limited in size, requiring careful use of augmentation and transfer learning.

8.6 Learnings from Week 4

This assignment provided exposure to **real clinical-grade AI challenges**, beyond standard datasets. Key learnings include:

- Understanding histopathology data characteristics
- Designing end-to-end object detection pipelines
- Handling annotation formats and dataset conversions
- Appreciating the complexity of small-object detection
- Recognizing ethical and deployment challenges in healthcare AI

This task bridged the gap between theoretical computer vision concepts and practical healthcare AI deployment.

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9. Challenges and Ethical Considerations

Healthcare AI faces unique challenges:

- Limited availability of labeled medical data
- Class imbalance in datasets
- Risk of overfitting
- Variability across imaging devices

- Ethical concerns regarding bias, fairness, and accountability

AI models in healthcare must be designed responsibly, ensuring transparency, reliability, and clinician oversight.

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10. Overall Learnings from the WiDS 5.0 Project

This project provided a comprehensive learning experience that bridged foundational AI concepts with real-world healthcare applications. The learnings progressed systematically from theory to practical deployment-oriented challenges.

10.1 Conceptual Learnings

- Gained a strong understanding of how **AI is applied in healthcare**, particularly in medical imaging and diagnostic workflows.
- Learned the **structure and purpose of healthcare and medical imaging reports**, enabling better contextual understanding of clinical data.
- Understood why **medical imaging problems differ fundamentally from general computer vision tasks**, especially in terms of data quality, scale, and responsibility.

10.2 Image Processing and Computer Vision Learnings

- Developed a solid foundation in **digital image processing**, including:
    - Image representation and pixel-level operations
    - Grayscale conversion and thresholding
    - Image resizing and interpolation techniques
    - Color space transformations (RGB vs HSV)

- Understood how **preprocessing decisions directly influence model performance**, particularly in medical imaging tasks where subtle details are clinically important.

10.3 Deep Learning and Model Architecture Learnings

- Gained theoretical understanding of **Convolutional Neural Networks (CNNs)** and their role in feature extraction from images.
- Learned the significance of **ImageNet and transfer learning** in training deep models with limited medical data.
- Studied major CNN architectures such as AlexNet, VGG, ResNet, and EfficientNet and their design motivations.

10.4 Practical Implementation Learnings

- Acquired hands-on experience with the **YOLO framework**, including different modes and tasks.
- Implemented **YOLOv8 for both image classification and object detection**, understanding their differences and use cases.
- Designed and executed a **complete end-to-end training pipeline**, including:
  - Dataset preparation
  - Annotation format conversion (COCO to YOLO)
  - Preprocessing and augmentation
  - Model training and inference

10.5 Healthcare-Specific Learnings

- Understood the **challenges of histopathology image analysis**, such as:
  - Small object size (lymphocytes)
  - Dense cell distribution
  - Stain variability across slides
  - Class imbalance between cell types
- Learned why **patch-based learning** is necessary for whole-slide image analysis.
- Recognized the importance of **model robustness, consistency, and reproducibility** in healthcare settings.

10.6 Ethical and Professional Learnings

- Gained awareness of **ethical considerations in healthcare AI**, including bias, fairness, accountability, and patient safety.
- Understood that AI models should function as **decision-support tools**, not replacements for clinicians.
- Developed a mindset focused on **responsible AI deployment**, especially in high-stakes domains like healthcare.

10.7 Skill Development

Through this project, I developed:

- Practical deep learning and computer vision skills
- Experience handling real-world medical datasets
- Debugging and problem-solving abilities in ML pipelines
- Improved technical documentation and reporting skills
- Confidence to pursue advanced AI-for-healthcare projects

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11. Future Scope

While this project focused on foundational concepts and a core detection task, there are several meaningful directions for future extension and improvement:

11.1 Full Automated TIL Scoring

- Extend the current lymphocyte and plasma cell detection model to:
    - Segment tumor and stroma regions
    - Compute quantitative TIL scores as defined in clinical guidelines
- This would fully align the pipeline with the **TIGER Grand Challenge objectives**.

11.2 Whole-Slide Image Analysis

- Move from patch-based learning to **whole-slide image inference** using:
    - Sliding window techniques
    - Multi-scale detection
- This would enable direct clinical applicability.

11.3 Model Performance Improvements

- Experiment with:
    - Larger YOLOv8 variants
    - Advanced data augmentation strategies
    - Class balancing techniques
    - Hyperparameter tuning
- Improve detection accuracy for small and rare cell types.

11.4 Explainable AI in Histopathology

- Integrate explainability methods such as:
    - Grad-CAM
    - Attention maps
- Help pathologists understand **why** a model makes certain predictions, increasing trust and adoption.

11.5 Cross-Dataset Generalization

- Evaluate model performance across:
    - Multiple hospitals
    - Different staining protocols
- Improve generalizability and robustness.

11.6 Clinical Validation and Deployment

- Collaborate with medical professionals for:
    - Clinical validation
    - Feedback-driven model refinement
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- Explore deployment as a **decision-support system** in pathology workflows.

11.7 Broader Healthcare Applications

- Apply similar pipelines to:
    - Other cancer types
    - Cell detection tasks
    - Medical image segmentation problems
- Extend learnings beyond breast cancer histopathology.

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